1. What is the α_{DC} when $I_C = 8.23$ mA and $I_E = 8.69$ mA?
2. A certain transistor has an $I_C = 25$ mA and an $I_B = 200\ \mu$A. Determine the β_{DC}.
3. What is the β_{DC} of a transistor if $I_C = 20.3$ mA and $I_E = 20.5$ mA?
4. What is the α_{DC} if $I_C = 5.35$ mA and $I_B = 50\ \mu$A?
5. A certain transistor exhibits an α_{DC} of 0.96. Determine I_C when $I_E = 9.35$ mA.

6. Determine the I_B, I_C, I_E, V_{BE}, V_{CE}, and V_{CB} in the circuit in Figure 1. The transistor has $\beta_{DC} = 150$.

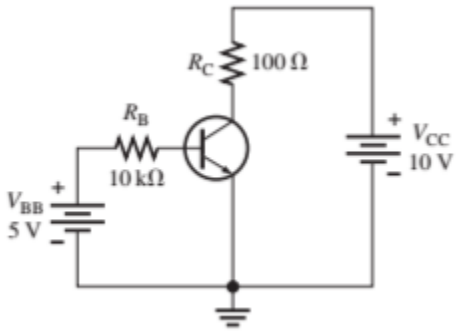


Figure 1

7. Determine whether or not the transistor is in saturation. Assume VCE = 0.2V.

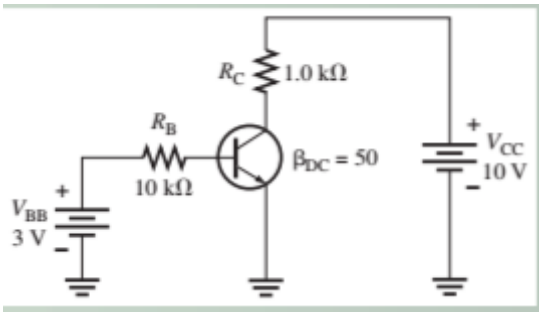


Figure 2